

Práctica uso de Docker

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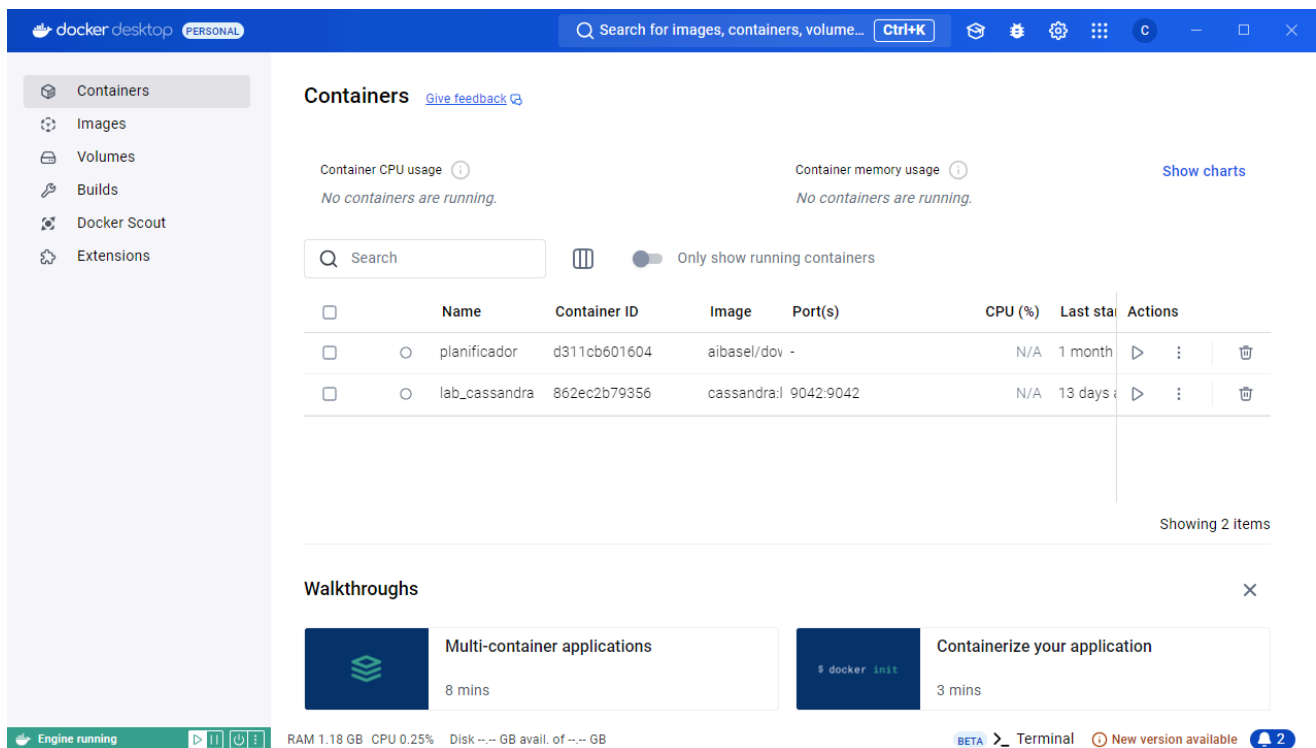
CU: 35-5267

Objetivos

Esta actividad te permitirá conocer los comandos básicos de Docker. Ponte a prueba ejecutando varios comandos sobre una instalación previa de Docker Desktop.

Parte 1. Ejercicios Guiados

Para realizar esta actividad, se asume que tienes instalado Docker Desktop en su versión reciente.



Instrucciones

Debe abrir la línea de comandos y ejecutar los comandos indicados, luego de ejecutar debe sacar una captura de pantalla y pegar debajo del comando

Primeros pasos

1. Comprueba la versión de Docker:

> `docker --version`

```
❏ ~ docker --version
Docker version 28.3.3, build 980b856816
```

2. Descarga una imagen de Docker, por ejemplo, httpd (Apache):

> `docker pull httpd`

```
❏ ~ docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
Digest: sha256:3198c1839e1a875f8b83803083758a7635f1ae999f0601f30f2f3b8ce2ac99e3
Status: Image is up to date for httpd:latest
docker.io/library/httpd:latest
```

3. Consulta las imágenes descargadas:

> `docker images`

```
❏ ~ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
httpd	latest	199e3a035264	12 days ago	117MB
postgres	16	5d77af46c4cb	13 days ago	452MB
node	20-alpine	7cdef5a33192	5 weeks ago	135MB
postgres	15	af3ea6376a7b	2 months ago	430MB
mongo	4.4	d896c071ac69	17 months ago	427MB

4. Ejecuta una instancia (contenedor) de la imagen descargada (httpd):

> `docker run -it -d -p 8080:80 httpd`

```
❏ ~ docker run -it -d -p 8080:80 httpd
758cdc298ab31f8a5490d5c906aa954e35886af82dd15d5a557e02c5149f9210
```

5. Consulta todos los contenedores creados:

> `docker ps`

```
❏ ~ docker ps
```

CONTAINER ID	IMAGE	COMMAND NAMES	CREATED	STATUS	PORTS
758cdc298ab3	httpd	"httpd-foreground"	27 seconds ago	Up 26 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

> `docker ps -a`

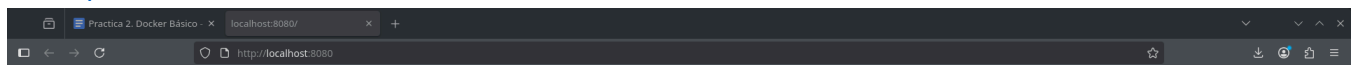
```

docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
758cdc298ab3   httpd      "httpd-foreground"      57 seconds ago   Up 56 seconds
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
0681fbc53274   node:20-alpine "docker-entrypoint.s..." 52 minutes ago   Exited (0) 52 m
inutes ago
b79efe1bdfaa   node:20-alpine "docker-entrypoint.s..." 53 minutes ago   Exited (1) 53 m
inutes ago
a95cc07925e9   httpd      "httpd-foreground"      About an hour ago Exited (0) 59 m
inutes ago
949ebbc73d77   postgres:16 "docker-entrypoint.s..." 6 days ago       Exited (0) 52 m
inutes ago
4182b4320c5c   postgres:16 "docker-entrypoint.s..." 7 days ago       Exited (0) 52 m
inutes ago
postgres-agenda

```

6. Comprueba el funcionamiento del contenedor desde el host:

> <http://localhost:8080>



It works!

7. Ejecuta la terminal de Linux dentro del contenedor:

> ls

```

root@758cdc298ab3:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  htdocs  icons  include  logs  modules
root@758cdc298ab3:/usr/local/apache2#

```

> docker exec -it 2e50e06a5c84

```

root@758cdc298ab3:/usr/local/apache2#

```

8. Borra el contenedor creado:

- > docker rm CONTAINER_ID
- > docker stop 2e50e06a5c84 *(para el contenedor antes de borrarlo)*
- > docker rm 2e50e06a5c84

```

❯ docker stop 758cdc298ab3
758cdc298ab3
❯ docker rm 758cdc298ab3
758cdc298ab3

```

9. Comprueba que el contenedor ha sido eliminado:

- > docker ps -a

```

❯ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
0681fbc53274	node:20-alpine	"docker-entrypoint.s..."	56 minutes ago	Exited (0) 56 minu
b79efe1bdfaa	node:20-alpine	"docker-entrypoint.s..."	56 minutes ago	Exited (1) 56 minu
a95cc07925e9	httpd	"httpd-foreground"	2 hours ago	Exited (0) About a
949ebbc73d77	postgres:16	"docker-entrypoint.s..."	6 days ago	Exited (0) 56 minu
4182b4320c5c	postgres:16	"docker-entrypoint.s..."	7 days ago	Exited (0) 55 minu

10. Borra la imagen usando la identificación del Comprueba que el contenedor ha sido eliminado:

- > docker rmi IMAGE_ID
- > docker images
- > docker rmi a8ea074f4566
- > docker rmi -f a8ea074f4566 *(forzar el borrado)*

```

❯ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
httpd latest 199e3a035264 12 days ago 117MB
postgres 16 5d77af46c4cb 13 days ago 452MB
node 20-alpine 7cdef5a33192 5 weeks ago 135MB
postgres 15 af3ea6376a7b 2 months ago 430MB
mongo 4.4 d896c071ac69 17 months ago 427MB
❯ docker rmi 199e3a035264
Untagged: httpd:latest
Untagged: httpd@sha256:3198c1839e1a875f8b83803083758a7635f1ae999f0601f30f2f3b8ce2ac99e3
Deleted: sha256:199e3a0352645ad4ae027e853af5986bd5b4766c3a5e98987ad10d5922eb7465
Deleted: sha256:00573b19d7727f02c759d09e37596d7790e794d8044bbda01c82e4ef255a5a46
Deleted: sha256:84e29bd3c0d55e14757bab091c4081f900a13e91d88ebe65d7711411ac757385
Deleted: sha256:d75c7aed0111141a372ee22ba367d8b7d6dfc111abd79544592065f98a57802c
Deleted: sha256:d39e6d1ae882eb3ae39fb74a830df8d28fcd5032b8eb94da68c5f488c51ece5d
Deleted: sha256:7def08ab353787eaf7f6c76df7a4de80178cd031f0b2e5267291b645d148600a

```

Borrado Forzado:

```

❏ ~ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
httpd         latest    199e3a035264   12 days ago    117MB
postgres      16        5d77af46c4cb   13 days ago    452MB
node          20-alpine 7cdef5a33192   5 weeks ago    135MB
postgres      15        af3ea6376a7b   2 months ago   430MB
mongo         4.4       d896c071ac69   17 months ago  427MB
❏ ~ docker rmi -f 199e3a035264
Untagged: httpd:latest
Untagged: httpd@sha256:3198c1839e1a875f8b83803083758a7635f1ae999f0601f30f2f3b8ce2ac99e3
Deleted: sha256:199e3a0352645ad4ae027e853af5986bd5b4766c3a5e98987ad10d5922eb7465
Deleted: sha256:00573b19d7727f02c759d09e37596d7790e794d8044bbda01c82e4ef255a5a46
Deleted: sha256:84e29bd3c0d55e14757bab091c4081f900a13e91d88ebe65d7711411ac757385
Deleted: sha256:d75c7aed0111141a372ee22ba367d8b7d6dfc111abd79544592065f98a57802c
Deleted: sha256:d39e6d1ae882eb3ae39fb74a830df8d28fcd5032b8eb94da68c5f488c51ece5d
Deleted: sha256:7def08ab353787eaf7f6c76df7a4de80178cd031f0b2e5267291b645d148600a

```

11. Reinicia el contenedor:

> docker restart CONTAINER_ID

```

❏ ~ docker ps
CONTAINER ID   IMAGE    COMMAND                  CREATED        STATUS        PORTS
cd0275342b13   httpd    "httpd-foreground"      4 seconds ago Up 4 seconds  0.0.0.0:8080->80/tcp
, [::]:8080->80/tcp priceless_ptolemy
❏ ~ docker restart cd0275342b13
cd0275342b13

```

> docker ps -a

> docker restart 2e50e06a5c84

```

❏ ~ docker ps -a
CONTAINER ID   PORTS          IMAGE    COMMAND                  CREATED        STATUS
cd0275342b13   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp httpd    "httpd-foreground"      31 seconds ago Up 18 seconds
0681fbc53274   node:20-alpine "docker-entrypoint.s..." About an hour ago Exited (0) About an
hour ago confident_turing
b79efe1bdfaa   node:20-alpine "docker-entrypoint.s..." About an hour ago Exited (1) About an
hour ago musing_archimedes
949ebbc73d77   postgres:16    "docker-entrypoint.s..." 6 days ago      Exited (0) About an
hour ago postgres-agenda-typeorm
4182b4320c5c   postgres:16    "docker-entrypoint.s..." 7 days ago      Exited (0) About an
hour ago postgres-agenda

```

12. Detén y levanta el contenedor:

> docker stop CONTAINER_ID

> docker stop 2e50e06a5c84

> docker ps *(comprueba que está parado el contenedor)*

> docker start 2e50e06a5c84

> docker ps

```

❏ ~ docker stop cd0275342b13 ✓
cd0275342b13
❏ ~ docker ps ✓
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
❏ ~ docker start cd0275342b13 ✓
cd0275342b13
❏ ~ docker ps ✓
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
cd0275342b13   httpd     "httpd-foreground"      About a minute ago   Up 1 second   0.0.0.0:8080->80
/tcp, [::]:8080->80/tcp   priceless_ptolemy

```

13. Mata el contenedor:

> docker kill 2e50e06a5c84

> docker ps

```

❏ ~ docker kill cd0275342b13 ✓
cd0275342b13
❏ ~ docker ps ✓
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
❏ ~ docker ps -a ✓
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
cd0275342b13   httpd     "httpd-foreground"      2 minutes ago   Exited (137) 5 seconds ago   priceless_ptolemy
0681fbc53274   node:20-alpine "docker-entrypoint.s..." About an hour ago   Exited (0) About an hour ago   confident_turing
b79efe1bdfaa   node:20-alpine "docker-entrypoint.s..." About an hour ago   Exited (1) About an hour ago   musing_archimedes
949ebbc73d77   postgres:16 "docker-entrypoint.s..." 6 days ago       Exited (0) About an hour ago   postgres-agenda-typeorm
4182b4320c5c   postgres:16 "docker-entrypoint.s..." 7 days ago       Exited (0) About an hour ago   postgres-agenda

```

14. Visualiza los detalles de la red definida con los clústeres:

> docker network

> docker network ls

> docker ps

```

❏ ~ docker network ls
NETWORK ID      NAME                                DRIVER  SCOPE
b6e2b285aded    bridge                             bridge  local
5a2852b74f99    crud-agenda-typeorm_default        bridge  local
2ae75d013454    crud-agenda_default                bridge  local
60ec37b87f98    host                               host    local
496b737cae44    lms-nestjs_pgnetwork               bridge  local
9f92d5950afd    none                                null    local
❏ ~ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES

```

15. Obtén información detallada sobre los recursos de Docker:

> docker Info

```
❏ 🏠 ~ docker info
Client:
Version:      28.3.3
Context:      default
Debug Mode:   false
Plugins:
  compose: Docker Compose (Docker Inc.)
    Version:  2.39.2
    Path:     /usr/lib/docker/cli-plugins/docker-compose

Server:
Containers: 5
  Running: 0
  Paused: 0
  Stopped: 5
Images: 5
Server Version: 28.3.3
Storage Driver: overlay2
  Backing Filesystem: btrfs
  Supports d_type: true
  Using metacopy: true
  Native Overlay Diff: false
  userxattr: false
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
  /var/run/cdi
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: 75cb2b7193e4e490e9fbdc236c0e811ccaba3376.m
runc version:
init version: de40ad0
Security Options:
  seccomp
  Profile: builtin
  cgroupns
Kernel Version: 6.12.41-1-MANJARO
Operating System: Manjaro Linux
```

```

OSType: linux
Architecture: x86_64
CPUs: 16
Total Memory: 15.29GiB
Name: w33-mau
ID: 8488d47c-697c-43c8-9175-20e518263215
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
::1/128
127.0.0.0/8
Live Restore Enabled: false

```

16. Revisa el historial de una imagen:

> docker history httpd

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
199e3a035264	12 days ago	CMD ["httpd-foreground"]	0B	buildkit.dockerfile.v0
<missing>	12 days ago	EXPOSE map[80/tcp:{}]	0B	buildkit.dockerfile.v0
<missing>	12 days ago	COPY httpd-foreground /usr/local/bin/ # buil...	138B	buildkit.dockerfile.v0
<missing>	12 days ago	STOPSIGNAL SIGWINCH	0B	buildkit.dockerfile.v0
<missing>	12 days ago	RUN /bin/sh -c set -eux; savedAptMark="\$(a...	32.5MB	buildkit.dockerfile.v0
<missing>	12 days ago	ENV HTTPD_PATCHES=	0B	buildkit.dockerfile.v0
<missing>	12 days ago	ENV HTTPD_SHA256=58b8be97d9940ec17f7656c0c6b...	0B	buildkit.dockerfile.v0
<missing>	12 days ago	ENV HTTPD_VERSION=2.4.65	0B	buildkit.dockerfile.v0
<missing>	12 days ago	RUN /bin/sh -c set -eux; apt-get install --...	5.63MB	buildkit.dockerfile.v0
<missing>	12 days ago	WORKDIR /usr/local/apache2	0B	buildkit.dockerfile.v0
<missing>	12 days ago	RUN /bin/sh -c mkdir -p "\$HTTPD_PREFIX" && ...	0B	buildkit.dockerfile.v0
<missing>	12 days ago	ENV PATH=/usr/local/apache2/bin:/usr/local/s...	0B	buildkit.dockerfile.v0
<missing>	12 days ago	ENV HTTPD_PREFIX=/usr/local/apache2	0B	buildkit.dockerfile.v0
<missing>	12 days ago	# debian.sh --arch 'amd64' out/ 'trixie' '@1...	78.6MB	debuerreotype 0.15

17. Busca una imagen según el nombre:

> docker search mongo

NAME	DESCRIPTION	STARS	OFFICIAL
mongo	MongoDB document databases provide high avai...	10617	[OK]
circleci/mongo	CircleCI images for MongoDB	16	
corpusops/mongo	https://github.com/corpusops/docker-images/	0	
rootpublic/mongo		0	
uselagoon/mongo		0	
ekesken/mongo	docker image for mongo that is configurable ...	1	
vepo/mongo	Mongo image with default settings	1	
litmuschaos/mongo		1	
jelastic/mongo	An image of the MongoDB database server main...	0	
arm64v8/mongo	MongoDB document databases provide high avai...	36	
zinobe/mongo	mongo with custom user	0	
s390x/mongo	MongoDB document databases provide high avai...	1	
kubedb/mongo		0	
cescoferraro/mongo	docker alpine mongo	0	
ccitest/mongo	CircleCI test images for Mongo	0	
amd64/mongo	MongoDB document databases provide high avai...	0	
sacashgit/mongo		0	
excellalabs/mongo	Dockerfile and scripts to setup a production...	2	
winamd64/mongo	MongoDB document databases provide high avai...	1	
mongodb/mongodb-community-server	The Official MongoDB Community Server	164	
healthcheck/mongo	https://github.com/docker-library/healthchec...	5	
pagarme/mongo		0	
pastvu/mongo		0	
sscpac/mongo	alpine mongo	0	
mongodb/mongodb-atlas-local	Create, manage, and automate MongoDB Atlas L...	8	